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Q1. Harshad number

ANSWER

```
n=int(input("enter the number"))
s=sum(int(i) for i in str(n))
if n%s==0:
    print("harshad number")
else:
    print("not a harshad number:")
for i in range(1,501):
    if i%sum(int(j) for j in str(i))==0:
        print(i,end=" ")
```

Q2. GCD

CODE

```
a=int(input("enter first:\n"))
b=int(input("enter second:\n"))
while b!=0:
    temp=a
    b=a
    a=temp%a
print("GCD IS",a)
```

INPUT

12
18

OUTPUT

enter first:
enter second:
GCD IS 12**Q3. sum to a target value**

ANSWER

```
def find_pairs(lst,target):
    seen=set()
    pairs=[]
    for num in lst:
        complement=target-num
        if complement in seen:
            pairs.append((complement,num))
        seen.add(num)
    return pairs
data=[int(x) for x in input("enter the number:").split()]
t=int(input("target:"))
result=find_pairs(data,t)
print("pairs",result)
```

Q4. Bitwise Operators

ANSWER

```
a=13(int)
b=21(int)
c=26(int)
result=31(int)
output:
a=13, b=21, c=26, result=31
0b111110037 0x1f
7 62 -32
data types:
aint
bint
cint
resultint
```

Q5. binary search

CODE

```
def rotated_binary_search(arr,target):
    low=0
    high=len(arr)-1
    while low<=high:
        mid=(low+high)//2
        if arr[mid]==target:
            return mid
        if arr[low]<arr[mid]:
            if target<= arr[low] and target<arr[mid]:
                high=mid-1
            else:
                low=mid+1
        else:
            if target>arr[mid] and target<=arr[high]:
                low=mid+1
            else:
                high=mid-1
    return -1
data=[4,5,6,7,0,1,2]
t=0
result=rotated_binary_search(data,t)
print("found at:",result)
```

OUTPUT

found at: 4